



Jamaica Constabulary Force

Standard Operating Procedures for Equipment Calibration

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1.0. Introduction

The Jamaica Constabulary Force (JCF) like so many organizations depends greatly on the use of technology to ensure efficiency and effectiveness in its processes. Several of these technologies are equipment/instruments used by members in varying areas of work to support or enhance the completion of tasks. These include but are not limited to computers, servers, speed radars and intoxilyzers. The extent to which these equipment/instruments perform to the highest standards and maintain their capability as designed and as required is the extent to which their efficiency and effectiveness can be assured.

Calibration is one method used by the JCF to ensure that these equipment/instruments are performing optimally. The frequency of calibrations is highly dependent on the particular type of equipment/instrument; however when done properly it enables users and stakeholders to have confidence in the results that they measure, record and subsequently control.

Calibration of equipment/instruments used by the JCF is of vital importance since more often than not their use may result in persons being penalized or charged, especially in the cases of speed radar guns and intoxilyzers. Inaccuracies in the readings generated can lead to serious litigations against the organization.

The goal of calibration, therefore, is to minimise any measurement uncertainty by ensuring the accuracy of the equipment being tested for use.

2.0. Purpose

The purpose of this document is to outline the Standard Operating Procedure to guide calibration activities for specially identified equipment, instruments and tools used in the Jamaica Constabulary Force and set out the minimum requirements for these calibration procedures.

3.0 Scope

This Standard Operating Procedure applies to calibration activities conducted internally (in-house) or outsourced, and covers all equipment/Instrument and tools used for production, inspection, and testing in the Jamaica Constabulary Force

4.0 Acronyms and Abbreviation

- 4.1 AD** Active Directory
- 4.2 CAD** Computer Aided Dispatch
- 4.3 ICTD** Information and Communication Technology Division

- 4.3 **NTP** Network Time Protocol
- 4.4 **QA** Quality Assurance
- 4.5 **QMS** Quality Management System
- 4.6 **UTC** Coordinated Universal Time

5.0 Definition

5.1 Calibration

Is a procedure used to ensure that a measuring instrument is standardized. Essentially it is comparing reading on one piece of equipment or system, with another piece of equipment that has been calibrated and referenced to a known set of parameters.

5.2 Intoxilyzer

A device that is used to obtain and analyze a breath sample. It will detect the level of ethanol in a person's breath, which analyzes the molecules based on how they interact with light

5.3 Server

Is a physical computer device/machine or software which provides services to other computer programmes or users.

5.4 Speed Radar Gun

A device that is used to measure the speed of moving vehicles.

5.5 Tuning Fork

Is a two-pronged fork-shaped instrument used in many applications to produce a fixed tone. It is used to calibrate the speed radar guns used to track the speed of vehicles.

6.0 Roles and Responsibilities

- 6.1 Calibration activities for the Intoxilizer 8000 and speed radars used in the JCF by are led by the Quality Assurance (QA) department of Public Safety and Traffic Enforcement Branch (PSTEB).
- 6.2 Users of all JCF equipment/instruments shall support the calibration activity by checking the equipment /instruments status before use, reporting issues, and or taking the equipment for calibration before due dates.
- 6.3 Superintendent of Police in charge of Traffic Enforcement Division has responsibility for ensuring that proper systems are in place to ensure:
 - 6.3.1 Intoxilyzers are properly maintained and are calibrated before every use.

- 6.3.2 Speed Radar are calibrated before use daily.
- 6.3.3 personnel are trained before the issuance for use of intoxilyzer
- 6.4 Superintendent of Police in charge ICTD has the responsibility to ensure all Servers and Computers on the JCF Active Domain are calibrated before installation and after installation; constantly monitored by the System Administrators daily, also proactively scheduled maintenance
- 6.5 The Assistant Commissioner of Police in charge Technology Branch has oversight responsibilities to ensure ICT policies and standards are complied with

7.0 Calibration Procedures

7.1 Intoxilyzer 8000 Calibration

7.1.1 Personnel

Only members who have completed the training program in the use, care, and handling of the Intoxilyzer 8000 may perform this calibration.

7.1.2 Equipment

The intoxilyzer device has a built-in calibration process that must be initiated prior to each use of the instrument. This calibration process involves a series of self-test that is executed when the device is turned on.

7.1.2.1 Boot process: The built-in self-test/self-calibration include the following tests:

1. **Air test** (*ensure that the air quality is clean and free from air pollutants*)
2. **Ram test** (*ensure that the Random Access Memory (RAM) in the device is working*)
3. **Modem test** (*ensure that the modem work to allow access network if required*)
4. **Printer test** (*ensure that the printer is working to enable the printing of results*)
5. **Temperature test** (*ensure that the temperature in the space is adequately cool*)
6. **Calibration test** - Calibration test is complete when all tests are completed to ensure the device is booted and displays a “**Pass**” reading and a “**Ready**” reading is displayed indicating the readiness of the device for use.

7.1.2.2 Fail boot:

- i. After the device completes its self-tests and the calibration boot process, if there is a fault the display screen will show an “**error**” reading. The device should be turned off and then turned on again. If the “**error**” message remains, the device should be taken out of service
- ii. The SP Traffic Enforcement Division shall make every effort to have device repaired through the respective manufacturer.

7.2 Speed Radar Calibration

- 7.2.1 Each Speed Radar is equipped with a Tuning Fork that is specially calibrated for the device but it can be used on other similarly configured devices. The standard Measurement for the calibration test is 80km/hr.
- 7.2.2 The tuning fork emits a tone that is used to monitor the frequency and indicates when the respective speed radar gun is consistent with the specified standard
- 7.2.3 A calibration check must be conducted on each Speed Radar at the start of each shift
- 7.2.4 During the check the operator uses the respective tuning fork to test the accuracy of the reading on the speed radar. A solid object (e.g. woodblock) is used to knock the tuning fork which causes the tuning fork to vibrate and emit a tone.
- 7.2.5 The tuning fork is then placed in front of the speed gun radar and the operator simulates capturing the speed of a vehicle by pointing the radar at the tuning fork and reading is captured.
- 7.2.6 If the reading shows 80Km, this is an indication that the device is working properly
- 7.2.7 The Calibration register will then be completed by the operator by capturing the following information.
 - i. Date and Time
 - ii. Location
 - iii. Serial Number of Radar
 - iv. Tuning Fork Serial Number
 - v. Reading
 - vi. Status (work, not working)
 - vii. Name and details of Operator
 - viii. Name and details of Supervisor

7.2.8 Calibration self check for Tru-Cam Speed Radar

In addition to other speed radars in the JCF tool box is the TruCAM speed radar which is an all-in-one digital video camera/laser speed and ranging device. The TruCAM not only measures speed, but can calculate the travelling time and distance between two following vehicles and produces full action video and still image evidence, for both speeding and tailgating.

7.2.8.1 This speed radar is calibrated by doing internal self checks. When these checks are successfully completed, the Self Test Results Screen is displayed briefly, the display segments are lit and a series of tones are emitted indicating the device is fully calibrated and ready for use.

7.2.8.2 If any of the test fails, a system error message is displayed and the instrument will not operate.

7.3 Domain Computers and Servers Calibration

7.31 Computers that are currently on the JCF Domain are managed by the System Administrator through the Active Directory Domain Services (AD); hence the Date and Time, permissions and user access are synchronized using the AD. If the time on the server is not synced with the client machine, the users will be prevented from logging on to the client machine. This is a daily routine activity.

7.3.2. The time on the Active Directory Domain Server is kept synchronized by a Network Time Protocol (NTP) Server which uses Coordinated Universal Time (UTC).

7.4 Computer Aided Despatch software (CAD)

7.4.1 The date and time on the CAD is synchronized with the *Stratum 1 Clock*, which is the most accurate clock.

7.4.2 Monthly checks shall be conducted by the System Administrator of the ICTD to determine if there is any variance in the date and time (CAD and Stratum 1 Clock) and where any is found the CAD is adjusted accordingly.

8.0 Legislative Authority/guiding principles

8.1 ISO 9001:2015 standard

8.2 ICT Policy

8.3 Intoxilyzer 8000 Operators' Guide

8.4 Talon Traffic Safety Radar Operator's Manual

8.5 Road Traffic Act

9.0 Records

9.1 Speed Radar Calibration Register

9.2 Intoxilyzer Calibration Register

Appendix A

Calibration Checklist

Equipment	Calibration Checks	Method
Speed Radar	Daily	Tuning Fork is used to conduct Calibration Checks as per manufacturer's manual
Intoxilyzer	Before each use	Self-Calibration as per manufacturer's manual
Domain Computers	Automated check at each instance the user attempts to log on to the machine	The date and Time will be synchronized using the Domain Controller. The Domain Controller receives its time from a Windows Time server that is located at the Data Centre, which uses Universal Coordinated Time (UTC) as the referenced time. Based on group policies that are configured on the Domain Controller, the user cannot connect to the Client machine if there is a difference between the server time and the client.
Stand-alone Computers	Web Clock – time.curby.net (time Check)	The Date and Time is synchronized using the windows time server
App and Database Servers	The NTP server is checked daily as part of the System Maintenance	Servers are synchronized using a local NTP Server that resides at the Data Centre.

APPENDIX B

JCF/TECB/PR/SSs/0001/2022/F01

Public Safety & Traffic Enforcement Branch

16 Lower Elletson Road

Kingston 2

Speed Radar Calibration Register

User Name	Date	Serial #	Working Status	Measurement/ Reading	Action	Supervisor's Name

APPENDIX C

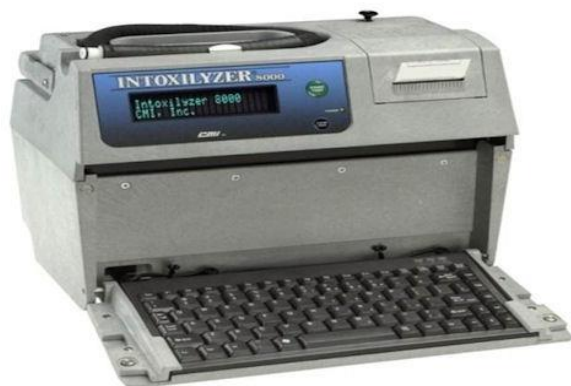
JCF/TECB/PR/SSs/0001/2022/F02

Public Safety & Traffic Enforcement Branch
16 Lower Elletson Road
Kingston 2

Intoxilyzer 8000 Calibration Register

User Name	Date	Serial #	Working Status	Measurement/Reading	Action	Investigating Officer

P1 Equipment (CAD, RADIO)	Monthly Checks are conducted to ensure check if there is any variance in the time	The time is synchronized by the Stratum 1 Clock, which is the most accurate clock.
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Intoxilyzer 8000



Speed Radar

